| Name: | Student ID: |
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OPTIMISATION PROBLEM SHEET

This sheet covers the three optimisation lectures. Question 1 looks at gradient descent and Newton's method on simple functions, where we can see exactly what happens. Questions 2 and 3 use Lagrange multipliers and duality. Question 4 asks you to check some of these results in Python. The early parts of each question are short; the later parts ask you to put the pieces together.

**1** Gradient descent with learning rate $r$ updates $w \leftarrow w - r f'(w)$ (or $\boldsymbol{w} \leftarrow \boldsymbol{w} - r\boldsymbol{\nabla} f(\boldsymbol{w})$ in several dimensions). In this question we see exactly when it works on quadratic functions.

(a) For $f(w) = \frac{1}{2}aw^2$ with $a > 0$, write down the gradient descent update and show that after $t$ steps $w_t = (1 - ra)^t w_0$. [3 marks]

$\overline{3}$

(b) For which learning rates does $w_t \to 0$ (the minimum)? Which learning rate is best? Describe what happens when $r = 1.5/a$ and when $r = 2.5/a$. [4 marks]

$\overline{4}$

(c) Now take $f(\boldsymbol{w}) = \frac{1}{2}(aw_1^2 + bw_2^2)$ with $a \geq b > 0$. Show that the two components evolve independently, and find the condition on $r$ for convergence. Show that the fixed learning rate that makes the slower of the two components converge fastest is $r = 2/(a+b)$, and that the error is then multiplied by $(\kappa - 1)/(\kappa + 1)$ each step, where $\kappa = a/b$ is the condition number. [5 marks]

5

(d) Take $a = 100$ and $b = 1$. What is the largest stable learning rate, and the best one? Roughly how many gradient descent steps are needed to reduce the error by a factor of $10^6$? How many steps would Newton's method need? [4 marks]

4

(e) Newton's method in one dimension updates $w \leftarrow w - f'(w)/f''(w)$. For $f(w) = \mathrm{e}^{w} - 2w$, show that the minimum is at $w^{*} = \ln 2$ and that Newton's update is $w \leftarrow w - 1 + 2\mathrm{e}^{-w}$. Starting from $w_0 = 0$, use a calculator to compute four iterations and the error $|w_t - \ln 2|$. What happens to the error? [4 marks]

$\frac{\quad}{4}$

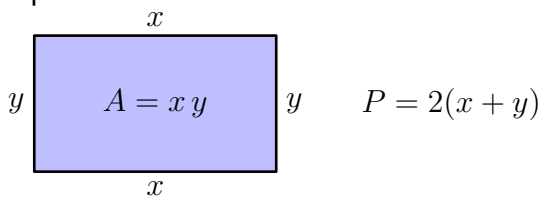
End of question 1

(a) $\frac{\quad}{3}$ (b) $\frac{\quad}{4}$ (c) $\frac{\quad}{5}$ (d) $\frac{\quad}{4}$ (e) $\frac{\quad}{4}$ Total $\frac{\quad}{20}$

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**2** To minimise (or maximise) $f(\boldsymbol{x})$ subject to $g(\boldsymbol{x}) = 0$ we form the Lagrangian $\mathcal{L}(\boldsymbol{x},\alpha) = f(\boldsymbol{x}) - \alpha g(\boldsymbol{x})$ and set its derivatives with respect to $\boldsymbol{x}$ and $\alpha$ to zero. For an inequality constraint $g(\boldsymbol{x}) \geq 0$ the KKT conditions add $\alpha \geq 0$ and $\alpha g(\boldsymbol{x}) = 0$.

(a) A rectangle has sides $x$ and $y$ and a fixed perimeter $2x+2y = P$. Use a Lagrange multiplier to find the sides that maximise its area $A = xy$. Show that $\alpha$ equals $\mathrm{d}A^*/\mathrm{d}P$, where $A^*$ is the maximum area: the Lagrange multiplier tells us how much the optimum improves if we relax the constraint.




$P = 2(x + y)$

[5 marks]

5

(b) Find the point $\boldsymbol{x}$ closest to the origin on the hyperplane $\boldsymbol{a}^\mathsf{T}\boldsymbol{x} = b$, by minimising $\|\boldsymbol{x}\|^2$ subject to this constraint. Hence show that the distance from the origin to the hyperplane is $|b|/\|\boldsymbol{a}\|$. (This is the geometry behind the margin of a support vector machine.) [6 marks]

6

(c) Minimise $f(x,y) = (x-2)^2 + (y-1)^2$ subject to the inequality $x + y \leq 2$. Write the constraint in the form $g(x,y) \geq 0$, decide whether it is active, and find the solution and the value of $\alpha$. [3 marks]

3

(d) Repeat with the constraint $x + y \leq 4$. [3 marks]

3

(e) Finally, take the *equality* constraint $x + y = 4$. Find the solution and $\alpha$. Why would a negative $\alpha$ be impossible for the inequality constraint $x + y \leq 4$? [3 marks]

3

End of question 2

| (a) $\frac{}{5}$ | (b) $\frac{}{6}$ | (c) $\frac{}{3}$ | (d) $\frac{}{3}$ | (e) $\frac{}{3}$ | Total $\frac{}{20}$ |
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**3** In the lectures we saw that the linear programme (the *primal*)

$$\min_{\boldsymbol{x}} \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x} \quad \text{subject to} \quad \mathbf{M}\boldsymbol{x} \geq \boldsymbol{b}, \quad \boldsymbol{x} \geq \mathbf{0}$$

has the *dual*

$$\max_{\boldsymbol{\alpha}} \boldsymbol{b}^{\mathsf{T}}\boldsymbol{\alpha} \quad \text{subject to} \quad \mathbf{M}^{\mathsf{T}}\boldsymbol{\alpha} \leq \boldsymbol{c}, \quad \boldsymbol{\alpha} \geq \mathbf{0}.$$

(a) Solve the linear programme: minimise $2x_1+3x_2$ subject to $x_1+x_2 \geq 4$, $x_1+3x_2 \geq 6$ and $x_1,x_2 \geq 0$. (The optimum of a linear programme is at a corner of the feasible region, so check the corners.) Write down the dual, solve it in the same way, and compare the two optimal values. [6 marks]

$\overline{6}$

(b) Show that for *any* $\boldsymbol{x}$ satisfying the primal constraints and *any* $\boldsymbol{\alpha}$ satisfying the dual constraints, $\boldsymbol{b}^{\mathsf{T}}\boldsymbol{\alpha} \leq \boldsymbol{c}^{\mathsf{T}}\boldsymbol{x}$. Why is this useful? [4 marks]

$\overline{4}$

(c) If the primal has $n$ variables and $k$ constraints (not counting $\boldsymbol{x} \geq \mathbf{0}$), how many variables and constraints does the dual have? Which would you rather solve when $n = 10^6$ and $k = 10$? [3 marks]

3

(d) Now consider the quadratic programme: minimise $\boldsymbol{x}^\mathsf{T}\boldsymbol{x}$ subject to $\mathbf{M}\boldsymbol{x} = \boldsymbol{b}$, where $\mathbf{M}$ is $k \times n$ with $k < n$. Form the Lagrangian, find $\boldsymbol{x}^*(\boldsymbol{\alpha})$, and show that the dual problem is

$$\max_{\boldsymbol{\alpha}} -\tfrac{1}{4}\boldsymbol{\alpha}^\mathsf{T}\mathbf{M}\mathbf{M}^\mathsf{T}\boldsymbol{\alpha} + \boldsymbol{\alpha}^\mathsf{T}\boldsymbol{b}.$$

Solve it, and show that $\boldsymbol{x}^* = \mathbf{M}^\mathsf{T}(\mathbf{M}\mathbf{M}^\mathsf{T})^{-1}\boldsymbol{b}$. How big is the matrix we have to invert? Check that $k = 1$ recovers question 2(b). [7 marks]

7

End of question 3

| (a) $\frac{}{6}$ | (b) $\frac{}{4}$ | (c) $\frac{}{3}$ | (d) $\frac{}{7}$ | Total $\frac{}{20}$ |
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**4** In this question you check some of these results in Python (or any other language). The following may be useful.

| | |
|---|---|
| `import numpy as np` | import the numpy module |
| `H = np.diag([100.0, 1.0])` | a diagonal matrix |
| `np.array([1.0, 1.0])` | a vector |
| `H @ w` | matrix–vector multiplication |
| `np.linalg.norm(w)` | the length $\|\boldsymbol{w}\|$ |
| `rng = np.random.default_rng(1)` | a random number generator with a fixed seed, so results can be repeated |
| `rng.integers(0, m, size=10)` | 10 random indices, for a mini-batch |

(a) Run gradient descent on $f(\boldsymbol{w}) = \frac{1}{2}(100w_1^2 + w_2^2)$ from $\boldsymbol{w} = (1,1)^\mathsf{T}$. For $r = 0.005$, $r = 2/101$ and $r = 0.0201$, count the steps until $\|\boldsymbol{w}\| < 10^{-6}$ (or report that it diverges). Compare with question 1. [5 marks]

5

(b) Add momentum, $\boldsymbol{v} \leftarrow \gamma\boldsymbol{v} - r\boldsymbol{\nabla} f(\boldsymbol{w})$, $\boldsymbol{w} \leftarrow \boldsymbol{w} + \boldsymbol{v}$. Count the steps for $\gamma = 0.9$, $r = 0.01$, and for the tuned values $r = 4/121$, $\gamma = (9/11)^2$. Compare with plain gradient descent. [5 marks]

5

(c) Code Newton's method for $f(w) = \mathrm{e}^w - 2w$ (question 1(e)) and print $\log_{10}|w_t - \ln 2|$ for the first six steps. [4 marks]

$\overline{4}$

(d) Generate 1000 data points with $x$ uniform on $[-1,1]$ and $y = 2x+1+0.5\epsilon$, where $\epsilon$ is a standard normal deviate. Fit $y \approx w_1 x + w_2$ by mini-batch SGD on the mean squared error, with batches of 10, $r = 0.1$ for 2000 steps and then $r = 0.01$ for 2000 steps. Record the loss on all the data after every step. Compare its average over the last 200 steps of each phase with the best possible loss, found by least squares. What does lowering $r$ do? [6 marks]

$\overline{6}$

End of question 4

| (a) $\overline{5}$ | (b) $\overline{5}$ | (c) $\overline{4}$ | (d) $\overline{6}$ | Total $\overline{20}$ |
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